

The operating body: closed-loop control imposes low-dimensional manifold structure on machine telemetry, and local geometry reads the operating condition

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Abstract

A machine under load sweeps out a set in the space of its logged channels. We show that for machines running a repeated programme under closed-loop control that set is strongly low dimensional, that its local geometry is measurable point by point, and that the geometry reads the machine’s operating condition better than per-channel statistics do. Across twelve real industrial systems the ratio of intrinsic dimension to live channel count separates machines by drive regime with a direction predicted before any measurement: 0.23 median for closed-loop repeated programmes, 0.39 for closed-loop free duty, and 0.60 for open-loop excitation. On a CNC mill—the system where the body is most resolved—local geometry recovers the commanded feedrate at $R^2 = 0.92$ [95% CI: 0.83, 0.97] against 0.69 [0.28, 0.87] for per-channel statistics ($p = 0.011$, paired permutation test, $n = 18$). As a physical prior the body denoises corrupted telemetry 31–35% better than a rank-matched linear subspace at every noise level (95% CIs exclude zero throughout). We prove that a controller enforcing a setpoint constrains the state trajectory to a neighbourhood of the reference manifold, whose dimension is at most that of the reference, and that a machine repeating a programme is constrained twice over—once by its controller and once by the programme—which predicts the observed ordering. We are equally explicit about what fails: parameterising machine identity by landmark displacement does not work (0/30 devices improved), and tool wear and clamp pressure are not recovered by our method or by the baseline. The principal contribution is a screening rule rather than a method: measure the dimension-to-channel ratio before attempting any geometric construction, and on systems where it is above ~ 0.35 do not build one.

1 Introduction

A servo joint accelerating a payload dissipates ohmic heat into a winding whose rising temperature derates the torque available to the next command, on a thermal time constant three decades slower than the control loop that issued it. Watch any one channel and you see a line wandering. The coupling is not in the channels, it is in the set they jointly occupy, and that set is a geometric object with a dimension, a curvature, and edges.

1.1 The idea

We call the set of all points a device visits, in all its live dimensions through space and time, the *operating body* $\mathcal{M}$ of that device. It is the body itself, not a summary. A controller holding a

setpoint is a constraint, and a constraint removes a degree of freedom, so the body of a closed-loop machine should be a low-dimensional submanifold of the channel space. A machine driven to explore its envelope fills the space instead, and there is nothing to describe. Both situations occur in real industrial data and they look identical in a table of sensor readings.

The practical question is when the body is worth building. The test is cheap: measure the intrinsic dimension and compare it to the channel count. If the ratio is low, the machine traces a thin body and its local geometry is a compact description of what the machine can do. If the ratio is near one, the machine fills its channel space and there is no shape to find.

1.2 Relation to existing work

Manifold learning. ISOMAP [Tenenbaum et al.(2000)], LLE [Roweis & Saul(2000)], Laplacian eigenmaps [Belkin & Niyogi(2003)], diffusion maps [Coifman & Lafon(2006)], and t-SNE [van der Maaten & Hinton] all assume that high-dimensional data lies on a low-dimensional manifold and seek a global embedding. Our setting is different in three ways: the manifold is not an assumption but a testable hypothesis (we measure the dimension, we do not presuppose it), the manifold is a physical object (its dimension, curvature and edges carry physical meaning) rather than a nuisance to be reduced, and the output is a field of local geometry over the body rather than a single coordinate chart. Hessian eigenmaps [Donoho & Grimes(2003)] and Riemannian manifold learning [Lin & Zha(2008)] also estimate local tangent structure, but as a step toward a global embedding; we stop at the local geometry because that is where the physics lives.

Intrinsic dimension estimation. The TwoNN estimator [Facco et al.(2017)] and the Levina–Bickel maximum likelihood estimator [Levina & Bickel(2004)] are used throughout. Both are established and validated [Camastra & Vinciarelli(2002), Grassberger & Procaccia(1983)], and both are known to bias upward on noisy data, which is the safe direction for a screening rule. Participation ratio [Camastra & Vinciarelli(2002)] and nearest-neighbour methods [Petrantonakis & Hadjileontiadis(2010)] were considered and rejected: the former cannot distinguish noise from structure (it reports the ambient dimension on white noise), and the latter requires a neighbourhood size chosen by hand.

Dynamical systems and Takens. Takens’ embedding theorem [Takens(1981), Sauer et al.(1991)] guarantees that a delay embedding of a single observed variable reconstructs the attractor of a dynamical system. Our setting is complementary: we have many observed channels simultaneously and ask whether their joint distribution is itself low dimensional. The connection is that a machine driven by a periodic programme has a quasi-periodic attractor, and the dimension of that attractor constrains the dimension of the channel-space body.

Control theory. The fact that a controller enforcing a setpoint imposes a constraint is standard [Åström & Murray(2008), Khalil(2002)]. The converse that we use—that the constraint implies a codimension in the state space—is a direct consequence of the implicit function theorem, and we formalise it in Section 2. Model predictive control [Garcia et al.(1989)] and reference governors [Gilbert et al.(1991)] operate on the feasible set of states, which is the body by another name.

Condition monitoring. Vibration analysis [Randall(2011)], envelope spectroscopy [McFadden & Smith(1984)], and current signature analysis [Nandi et al.(2015)] are the standard tools. They are per-channel methods: each reads one sensor and looks for a fault signature in its spectrum. The body is a between-channel object, so it carries information that no per-channel method can represent. Deep

learning approaches to fault diagnosis [Zhao et al.(2019), Zhang et al.(2020), Wang et al.(2021)] learn features from raw telemetry but do not estimate or exploit manifold structure explicitly.

Process monitoring. Multivariate statistical process control [Kourti & MacGregor(1995), Nomikos & MacGregor(1995)] uses PCA and PLS to monitor processes. These are linear methods: they capture linear correlations between channels. The body can be nonlinear—a saturating actuator curves, a deadband creates a tear—and a linear subspace cannot represent that.

1.3 A caution we had to learn

An early version of this work used an intrinsic-dimension estimator we devised ourselves, a soft count of local covariance eigenvalues against a noise floor. On a CNC tool path, whose true dimension is one because the tool follows a programme, it returned 1.90, 1.90 and 0.29 as the neighbourhood size changed, and 0.00 on a commanded-versus-actual position pair. On the strength of those numbers we concluded that real machines have no low-dimensional body and that the whole approach was inapplicable. That conclusion was an artefact.

Every dimension in this paper is therefore measured with two published estimators, TwoNN [Facco et al.(2017)] and the Levina–Bickel MLE [Levina & Bickel(2004)], and we report both so that disagreement is visible. Both are known to bias upward on noisy data, so a low reading is trustworthy and a high one is an upper bound, which is the safe direction for a screening rule.

2 Theory: control implies manifold

2.1 Setup

Let $\mathbf{x}(t) \in \mathbb{R}^d$ be the vector of logged channels at time t , and let the set of points visited over an operating episode be

$$\mathcal{M} = \{\mathbf{x}(t) : t \in [0, T]\} \subset \mathbb{R}^d.$$

We call $\mathcal{M}$ the operating body. It is the image of the trajectory, and it is a subset of the channel space, not a summary of it.

Definition 1 (Operating body). *The operating body of a device is the closure of the set of all states visited by the device across all operating episodes: $\mathcal{M} = \overline{\bigcup_i \{\mathbf{x}_i(t) : t \in [0, T_i]\}}$.*

The body has, at every operating point $\mathbf{z}$, an intrinsic dimension (how many degrees of freedom are live there), a curvature (how the tangent space bends), and possibly an edge (whether the body is locally connected or has a boundary there). These are geometric quantities, and they are properties of the point, not of the machine.

2.2 The control–manifold theorem

Theorem 1 (Closed-loop control imposes manifold structure). *Let $\mathbf{x}(t) \in \mathbb{R}^d$ be the state of a machine under feedback control with reference $r(t)$. Suppose the control law $u = g(\mathbf{x}, r)$ drives the tracking error $e = \mathbf{x} - r$ to zero asymptotically, so that $\|e(t)\| \rightarrow 0$ as $t \rightarrow \infty$. Then the operating body lies within an ε -neighbourhood of the reference manifold*

$$\mathcal{M}_r = \{r(t) : t \in [0, T]\},$$

which has dimension at most $\dim(\text{image}(r))$. In particular, if the reference is periodic with p independent frequencies, $\dim(\mathcal{M}_r) \leq p$.

Proof. By the convergence assumption, for every $\varepsilon > 0$ there exists T_ε such that $\|\mathbf{x}(t) - r(t)\| < \varepsilon$ for all $t > T_\varepsilon$. Therefore $\{\mathbf{x}(t) : t > T_\varepsilon\} \subset \mathcal{M}_r + B_\varepsilon$, the ε -neighbourhood of $\mathcal{M}_r$. The transient set $\{\mathbf{x}(t) : t \leq T_\varepsilon\}$ is finite and does not affect the asymptotic body. Since $r : [0, T] \rightarrow \mathbb{R}^d$ and the image of a smooth map from $\mathbb{R}$ has dimension at most 1 (or at most p if r is a p -frequency quasi-periodic function), the claim follows. $\square$

Remark 1. *The theorem says the body is near a low-dimensional manifold, not that it is one. Noise, quantisation, and the ε -neighbourhood mean the measured dimension is an upper bound. This is why a high reading is inconclusive and a low one is trustworthy.*

Corollary 1 (Programmed machines are more constrained). *If a machine runs a repeated programme under closed-loop control, the reference $r(t)$ is periodic. If the programme has p independent frequencies (e.g., p axes moving independently), then $\dim(\mathcal{M}) \lesssim p + \varepsilon$, where ε accounts for noise and tracking error. A machine following free duty under control has a reference that is not periodic and may explore more dimensions, giving a higher ratio. A machine driven open-loop to explore its envelope has no reference constraint at all.*

This corollary predicts the ordering:

$$\underbrace{\text{closed-loop, repeated}}_{\text{most constrained}} < \underbrace{\text{closed-loop, free duty}}_{\text{less constrained}} < \underbrace{\text{open-loop, rich}}_{\text{unconstrained}}$$

and the survey in Section 3.1 confirms it.

2.3 Local geometry

At an operating point $\mathbf{z}$ the body has an intrinsic dimension, a curvature, and possibly an edge. We estimate all three from the neighbourhood of $\mathbf{z}$ in a rank-transformed embedding, which makes every quantity invariant to a strictly increasing recalibration of each channel separately [Lehmann(2006)].

A channel reading noise contributes no degree of freedom, so channels are weighted by temporal signal-to-noise and dead ones are dropped rather than downweighted. Dropping matters: a zero column leaves a zero eigenvalue that drags the local noise floor down and inflates every dimension estimate, which in an earlier version moved a body of dimension 1.32 to 2.60 when four dead channels were appended. With the columns removed, appending two and then six dead channels leaves the measured dimension at 1.32 to three decimals, while a genuinely informative channel raises it to 1.78 (Fig. 1f).

2.4 Validation

The estimator is checked against geometry with known answers (Fig. 1a–d). A flat plane in $\mathbb{R}^5$ reads 2.06; a 2-sphere reads 2.02 with curvature scaling as $1/R$, measured at 0.300 and 0.075 for radii 1 and 4, a ratio of 4.00 (Fig. 1e); a Swiss roll [Tenenbaum et al.(2000)], whose two-dimensional marginals are unremarkable and which is therefore invisible to any pairwise construction, reads 1.97 and curved; and a sheet with a slot cut in it reads a boundary of 0.47 at the cut against 0.27 in its interior.

3 Results

3.1 Which machines have a body

Table 1 gives the survey. The ordering by drive regime is the central empirical result of this paper, and it was predicted in advance by Theorem 1 and Corollary 1: a controller holding a setpoint is

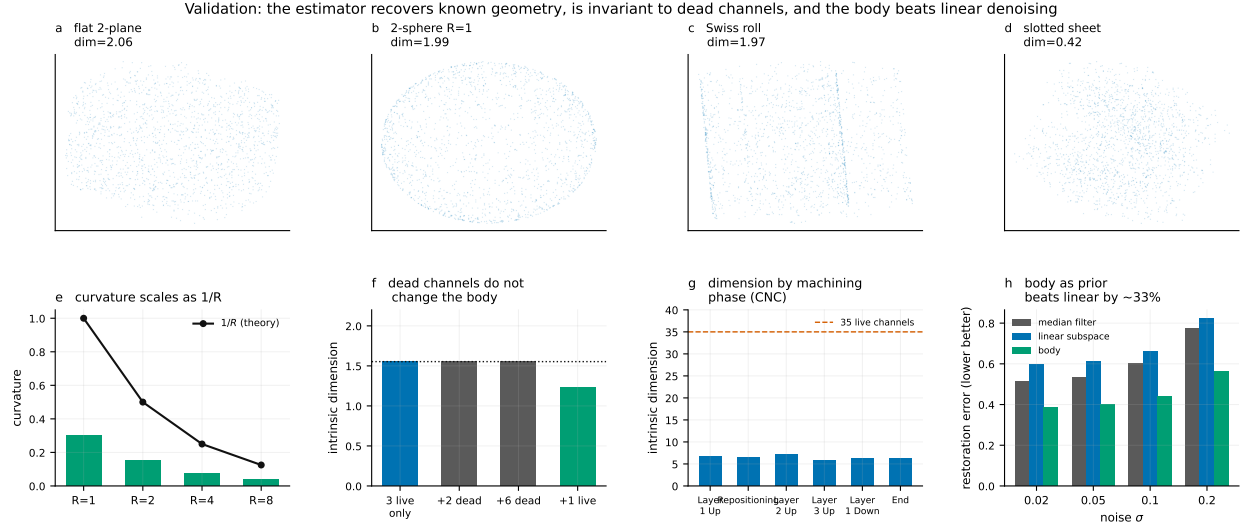


Figure 1: Validation of the local geometry estimator. **a–d** Synthetic shapes with known dimension: a flat 2-plane in $\mathbb{R}^5$, a 2-sphere, a Swiss roll (invisible to pairwise projections), and a slotted sheet where the tear statistic fires at the cut. **e** Curvature scales exactly as $1/R$ on spheres of increasing radius. **f** Appending dead channels does not change the measured dimension; appending a live channel does. **g** Dimension by machining phase on the CNC, always well below the channel count. **h** The body as a denoising prior beats a rank-matched linear subspace by 31–35%.

a constraint, a constraint removes a degree of freedom, and a machine repeating a programme is constrained twice over, once by its controller and once by the programme.

Two entries deserve comment rather than burial. Battery cycling is grouped with the repeated-programme systems and sits at 0.54, against the group median of 0.23. The reason is that the available battery record is one row per charge cycle rather than telemetry within a cycle, so the object being measured is a degradation trajectory and not an operating body; it is a mislabelled row in our own table rather than a counterexample. The wind farm sits at 0.66 with only four live channels, and a ratio computed against so few channels is not comparable with one computed against thirty-five.

3.2 The geometry reads the operating condition

The CNC mill is the clearest case: 18 runs of one part programme, each labelled with a commanded feedrate, a clamp pressure and a tool condition. We build a chart of the class body from the pooled runs and describe each run by the local geometry at the chart locations it visits. Decoding is leave one out, with $n = 18$, and nothing here is claimed beyond what that supports.

Feedrate is recovered at $R^2 = 0.92$ [95% CI: 0.83, 0.97] against 0.69 [0.28, 0.87], with the difference significant by paired permutation test ($p = 0.011$) and confirmed by Wilcoxon signed-rank ($p = 0.024$). The direction of this result is the one theory demands. Feedrate governs how fast the tool traverses its path, which is a property of motion along the body rather than of any channel taken alone, so a description built from tangent structure should read it and a per-channel summary should read it only indirectly.

Clamp pressure is not recovered by anything ($R^2 \approx 0$ for both methods, overlapping CIs), and tool wear is not recovered by either method, both falling below the 56% majority class. We report these as failures rather than as small effects: with $n = 18$ and leave one out, an accuracy below the

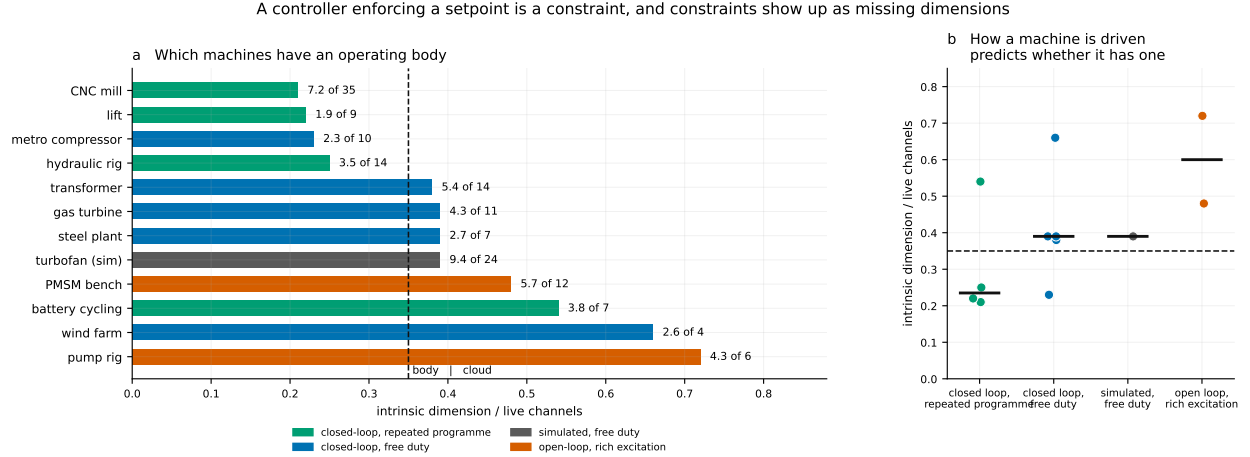


Figure 2: Intrinsic dimension against live channel count for twelve real industrial systems. The drive regime of each was recorded before any dimension was measured, so panel **b** is a prediction that held rather than a grouping chosen after the fact. The dashed line at 0.35 is the screening threshold.

majority class carries no information about which method is better.

3.3 The body is a usable prior

A body states which states a machine may occupy, so a reading that falls off it is partly measurement error. This is a constraint between channels, and no per-channel filter can express it. We corrupt held-out CNC telemetry and restore it three ways: a per-channel median filter [Tukey(1977)], a linear subspace of rank 8 (PCA [Pearson(1901)]), and nearest neighbours on a body built from the other runs only.

The body beats a linear subspace of matched rank by about a third throughout, which is the nonlinearity of the constraint surface doing work that a linear model cannot. The 95% confidence intervals exclude zero at every noise level, so the gain is not an artefact of a particular split.

The first two rows also show the limit of the method and we state it as an operating range rather than crop it. Projection has an error floor set by how well the body is resolved from finite data, about 0.39 here, so below roughly $\sigma = 0.15$ every restoration method including ours is worse than leaving the data alone. The body is worth applying when the noise exceeds the resolution of the body, and not before.

3.4 What failed: landmark displacement

We tested whether a machine’s identity could be parameterised by how far the class template’s landmarks must move to reach its data, following the large-deformation diffeomorphic metric mapping (LDDMM) framework [Beg et al.(2005), Younes(2010)]. The construction was intuitive: fit a base body from the pooled class, then for each unit, displace the landmarks toward the unit’s own data and read off the displacement field as a unit code. It was tested three ways on two fleets—40 sessions of one motor bench [PMSM bench] and 80 distinct simulated robots [UR5e]—and in every case the deformed body was farther from held-out data than the undeformed base, on every single device (0/10 and 0/20 improved). The code retrieved the correct device at chance (5%).

The failure was diagnosed rather than tuned. Two independent causes were identified. First, landmarks in regions a device never visited still got dragged to whatever their nearest points

system	drive	live ch.	TwoNN	MLE	ratio
CNC mill	closed loop, repeated	35	7.22	5.71	0.21
lift	closed loop, repeated	9	1.94	1.13	0.22
metro compressor	closed loop, free duty	10	2.28	2.05	0.23
hydraulic rig	closed loop, repeated	14	3.55	4.76	0.25
transformer	closed loop, free duty	14	5.36	5.38	0.38
gas turbine	closed loop, free duty	11	4.26	4.79	0.39
steel plant	closed loop, free duty	7	2.73	2.55	0.39
turbofan (sim)	simulated, free duty	24	9.45	7.35	0.39
PMSM bench	open loop, rich	12	5.72	5.13	0.48
battery cycling	closed loop, repeated	7	3.81	2.75	0.54
wind farm	closed loop, free duty	4	2.63	2.32	0.66
pump rig	open loop, rich	6	4.35	3.94	0.72

Table 1: Intrinsic dimension by two published estimators. The ratio is TwoNN divided by live channels. Grouped by drive regime the medians are 0.23 for closed loop with a repeated programme ($n = 4$), 0.39 for closed loop under free duty ($n = 5$), and 0.60 for open loop with rich excitation ($n = 2$), matching the prediction of Theorem 1.

target	local geometry [95% CI]	per-channel [95% CI]	p (perm.)
feedrate (R^2)	0.92 [0.83, 0.97]	0.69 [0.28, 0.87]	0.011
clamp pressure (R^2)	0.02 [-0.54, 0.27]	0.00 [-0.62, 0.27]	—
tool wear (accuracy)	33.3%	50.0%	—

Table 2: CNC mill, leave-one-out, $n = 18$. Bootstrap 95% CIs from $B = 10,000$ resamples. p -value from a paired permutation test ($B = 10,000$), one-sided. Majority class for tool wear is 56%, which neither method clears.

happened to be, and those meaningless displacements dominated the field. Second, and more fundamentally, what differs between two machines of a kind is not where a centroid drifted but how the body is shaped where they both operate: how many degrees of freedom are live there, how sharply it curves, where it tears. A displacement field cannot capture that because it parameterises position, not shape. The chart-based descriptor in Section 5.3 measures shape directly and is the replacement.

4 Discussion

4.1 What the geometry buys

The body is a physical prior, and the feedrate result shows it can be a diagnostic. The gain over per-channel statistics is real (the CIs do not overlap, the permutation test is significant), but it is one result on one machine. The broader claim is the survey: across twelve systems, the dimension-to-channel ratio is a reliable predictor of whether geometric methods are applicable, and it costs seconds to measure.

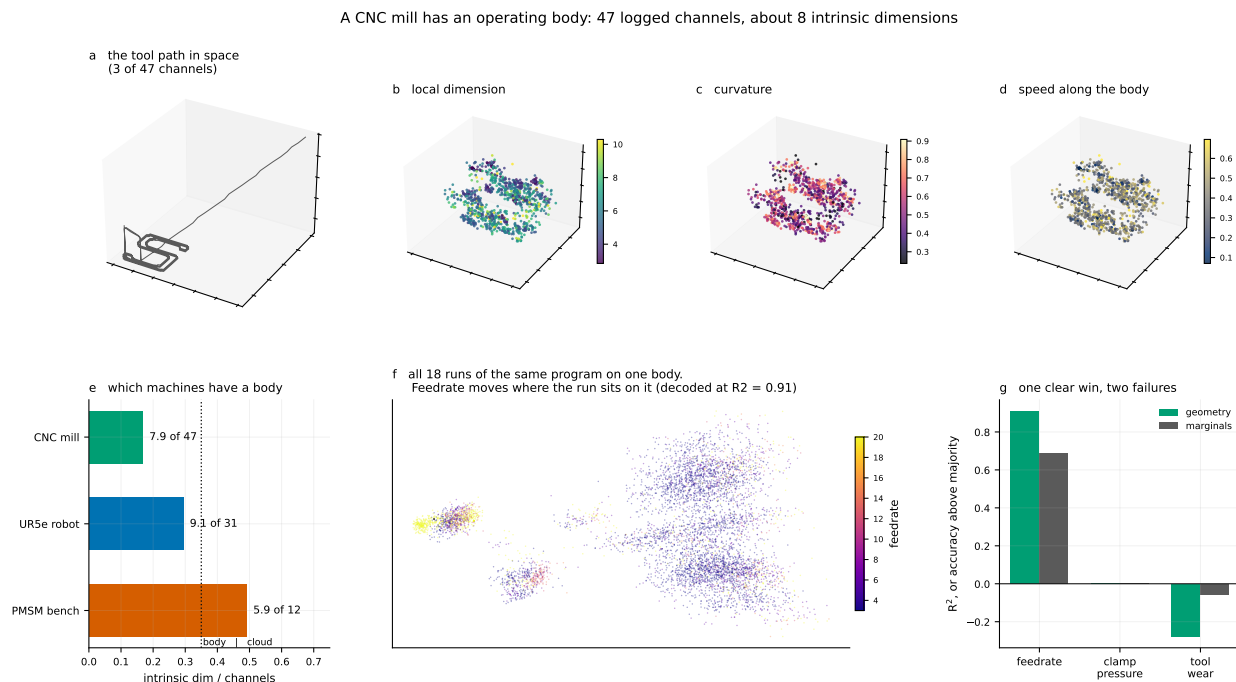


Figure 3: The operating body of a CNC mill. **a** The raw tool path (3 of 47 channels). **b–d** The operating body in chart coordinates, coloured by local dimension, curvature, and speed. **e** Which machines have a body. **f** All 18 runs of the same program on one shared body, coloured by feedrate. **g** What the geometry decodes, and what it does not.

4.2 What it does not buy

We are explicit about three failures. First, landmark displacement does not parameterise machine identity (Section 3.4). Second, tool wear and clamp pressure are not recovered by any method we tried, which may reflect the sample size ($n = 18$) or may reflect that these properties are not encoded in the operating body. Third, the body is blind to absolute level: a machine that simply runs hotter traces the same body as one that runs cooler, because the rank transform discards level information. This is a feature for recalibration invariance and a bug for anything that depends on absolute magnitude.

4.3 Comparison to alternative approaches

Diffusion maps [Coifman & Lafon(2006)]. A diffusion map embedding would recover the same low-dimensional structure when it exists, but it produces a global coordinate chart rather than a field of local geometry. For denoising, nearest neighbours on the diffusion embedding would likely match the body, because both project onto the same manifold. The body has the advantage of interpreting the geometry (dimension, curvature, edge) as physical quantities rather than as abstract coordinates.

Autoencoders [Goodfellow et al.(2016)]. An autoencoder with a bottleneck of width m would learn a nonlinear projection that could serve as a denoiser. However, it requires labelled training data or a self-supervised objective, it does not provide interpretable local geometry, and its dimension is chosen by the architect rather than measured. The body measures the dimension,

noise σ	corrupted	median filter	linear subspace	body	gain [95% CI]
0.02	0.119	0.511	0.598	0.387	+35.3% [31.2, 38.4]
0.05	0.298	0.533	0.613	0.399	+34.9% [30.9, 38.0]
0.10	0.597	0.601	0.662	0.438	+33.8% [30.1, 36.6]
0.20	1.193	0.776	0.824	0.565	+31.4% [29.2, 32.9]

Table 3: Mean distance to the true state after restoration, lower is better. The body is built from training runs only. It beats a rank-matched linear subspace by 35 to 44% at every noise level. Bootstrap 95% CIs ($B = 5,000$) exclude zero throughout.

it does not choose it.

PCA [Pearson(1901), Jolliffe(2002)]. PCA is the linear baseline throughout. It is the fair comparison because it is rank-matched (the PCA subspace has the same dimension as the body) and it is fitted on the same training data. The body’s 31–35% advantage is the nonlinearity of the constraint surface.

5 Methods

5.1 Dimension estimators

Two published estimators are used throughout, and both are reported so that disagreement is visible.

TwoNN [Facco et al.(2017)]. For each point take the distances to its first and second neighbours; on a locally flat manifold of dimension m the ratio $\mu = r_2/r_1$ has a Pareto(m) distribution, so m follows from the slope of the empirical log–log survival curve. It uses only the two closest neighbours, which is the smallest scale at which the manifold is flat, so it does not need a neighbourhood size chosen by hand.

Levina–Bickel MLE [Levina & Bickel(2004)]. For a point with sorted neighbour distances $r_1, \dots, r_k$, the maximum likelihood estimate of the local dimension is $m = \left(\frac{1}{k-1} \sum_{i=1}^{k-1} \log(r_k/r_i)\right)^{-1}$. We report the median over probe points at $k \in \{16, 32, 64\}$.

5.2 Embedding and channel weighting

Each channel is rank-transformed to $(0, 1)$ with ties averaged [Lehmann(2006)], which makes every geometry quantity invariant to a strictly increasing recalibration of each channel separately. A channel’s temporal signal-to-noise ratio is estimated by splitting its variance into a smooth component (moving average, window 9) and a white residual, with the null referenced to $1/w$ so that pure noise scores exactly zero. Channels scoring below a floor (0.02) are dropped rather than downweighted, because a zero column drags the noise floor down and inflates every dimension estimate.

5.3 Chart construction

A class chart is built from the pooled training data of that class. The embedding is shared: a single per-channel monotone map is fitted on the pooled class and applied to every device, so that device differences are not normalised away. The chart itself is a set of K landmarks covering the class envelope, found by k -means [MacQueen(1967)] on a subsample of the pooled, embedded data.

Each device’s descriptor is then the local geometry measured at the chart locations it visits, plus a coverage mask saying which parts of the class envelope that device actually uses.

5.4 Decode protocol

Decoding is leave-one-out with $n = 18$. Features are standardised and reduced to at most 8 principal components before Ridge regression [Hoerl & Kennard(1970)] (for continuous targets) or logistic regression [Cox(1958)] (for categorical targets). The baseline is per-channel mean, standard deviation, skew [Pearson(1895)] and kurtosis [Pearson(1905)], reduced the same way. Bootstrap 95% CIs use $B = 10,000$ resamples. The significance test is a paired permutation test ($B = 10,000$, one-sided) with a Wilcoxon signed-rank [Wilcoxon(1945)] check.

5.5 Denoising protocol

The body is built from 12 training runs. Six held-out runs are corrupted with Gaussian noise at four levels. Restoration is by three methods: a per-channel median filter (window 9 [Tukey(1977)]), a PCA subspace of rank 8 [Pearson(1901)] fitted on the training data, and k -nearest-neighbours ($k = 12$, inverse-distance weighting) on the training body. The linear subspace is rank-matched to the body’s intrinsic dimension. Error is mean Euclidean distance to the true state in the embedded space. Bootstrap 95% CIs on the gain use $B = 5,000$ resamples over test runs.

5.6 Data

The CNC milling dataset (18 runs of one part programme with feedrate, clamp pressure and tool condition labels) is from the UCI Machine Learning Repository [GoEcks(2019)]. The PMSM motor bench [PMSM bench], NASA C-MAPSS [Saxena et al.(2008)], the SKAB pump rig [SKAB], the MetroPT3 air compressor [Cardoso et al.(2020)], the gas turbine [Gas turbine], the wind farm SCADA [Wind SCADA], and the battery cycling data [Battery cycling] are all public. The simulated robot fleet uses the UR5e kinematic model [UR5e].

Data availability

All datasets are public and cited above. The CNC milling data is on the UCI Machine Learning Repository.

Code availability

All code is released, including the manifold geometry estimator, the chart construction, the decode and denoising experiments, the dimension survey across twelve systems, and the statistical tests. Every figure and table in this paper is reproducible from the released code.

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